

Physics-Informed Neural Networks for Ising-like Zimm-Bragg Dynamics

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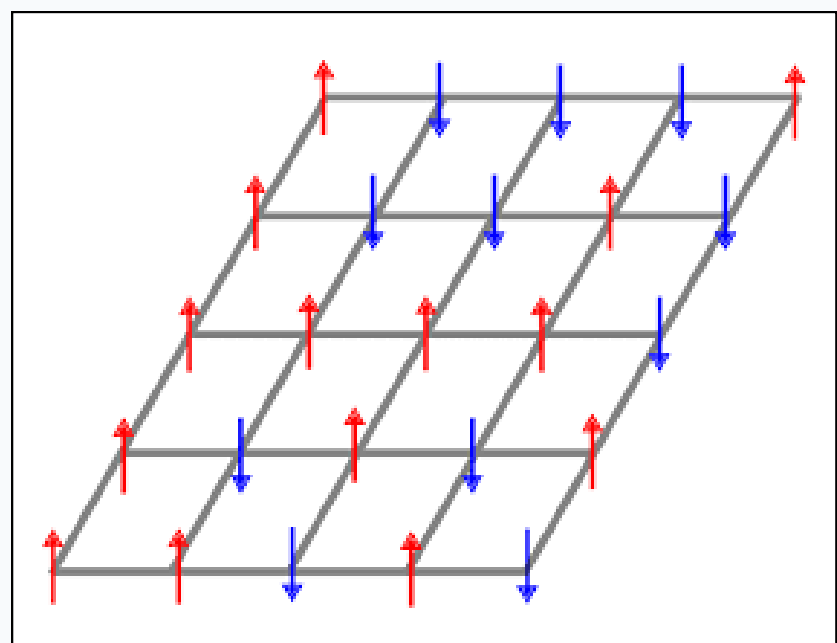
Introduction

Zimm-Bragg systems describe the cooperative transition of polypeptides between α -helical and random-coil states. Existing predictive approaches are split into:

- Monte Carlo, which is physically exact, but expensive at scale.
- Neural networks, which are fast, can violate the physics

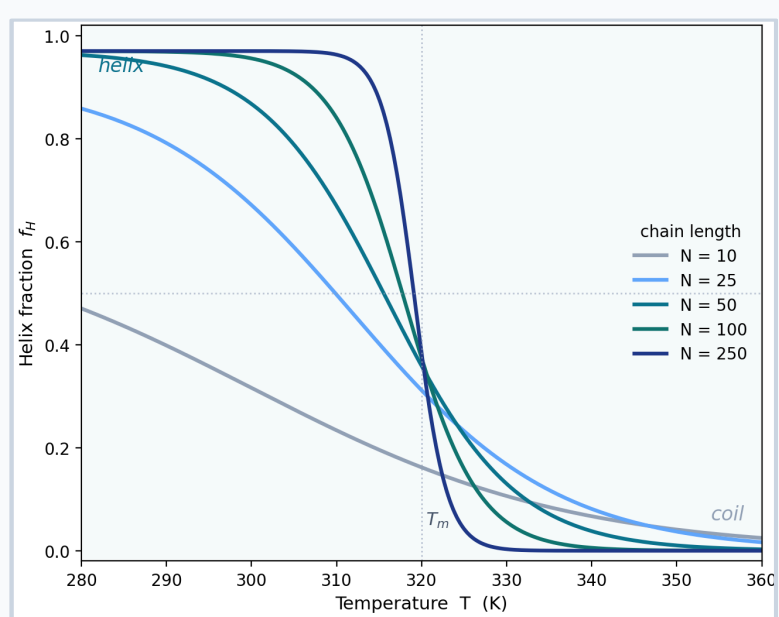
This work develops a hybrid framework that embeds the hydrodynamic approximation of the ZimmBragg model directly into a deep-learning loss, giving fast inference that respects the underlying physics

What is the Ising model?



A grid of binary spins (+1 or -1) that prefer to match their neighbors. Cool the system down and the spins align, but heat it up and they go random. Invented for magnets, the same recipe shows up wherever local interactions produce collective behavior, including the helix-coil transition.

What is the Zimm-Bragg model?



Zimm-Bragg is the version of the Ising chain built for protein folding. Each residue is either α -helix or random coil, and two parameters tune the chain: s , the preference for helix at a residue, and σ , the cost of starting a new helix region.

What are Glauber dynamics?

Glauber dynamics is a simple rule for how an Ising chain evolves in time. At each step you pick one spin at random and flip it with a probability that depends on the energy cost of the flip. Run the rule for long enough and the chain relaxes into the equilibrium the Ising model predicts.

What is a PINN?

A PINN is a neural network that gets graded on more than just fitting data. The training loss adds a second penalty for violating the physical equations of motion, so the model ends up both data-faithful and physics-faithful.

What are the Optimizers?

A total of 2 Optimizers were used for the model training:

- Adam: First-order gradient descent with momentum. Fast and exploratory, but imprecise near a minimum.
- L-BFGS: Quasi-Newton method that uses curvature from past gradients. Slow and demanding, but very decisive once close to a minimum.

Research Questions

- Can a physics-informed neural network learn the dynamics of a Zimm-Bragg chain from observed helicity data alone?
- Can it recover the underlying Glauber coefficients without ever being exposed to them?
- Does the physics term in the loss let the model generalize beyond its training window, and across the full parameter space, where a Standard Neural Network does not?

Key Numbers

- Data: 9900 Supervised Data Points
- Train/test split: 50/50 in time (extrapolation tests)
- Network: ~13K parameters (4 layers $\times$ 64 units MLP) plus 3 Glauber scalars
- Chain: 256 spins, periodic boundary
- Fit error: MSE $\sim 10^{-7}$ on $m(t)$, $\sim 10^{-5}$ on $\varepsilon(t)$, $\sim 10^{-7}$ on $\varepsilon_2(t)$

Methodology

- Averaging over the chain reduces its dynamics to 2 coupled ODEs (m and ε) with three Glauber coefficients
- We train a single PINN on the helicity data and recover the coefficients from the m-equation.
- Both $\hat{\varepsilon}$ and $\hat{\varepsilon}_2$ then follow analytically from the m-equation.
- Test predictive capabilities by extrapolation

Hydrodynamic Equations

Helicity

$$\dot{m} = -(1 - a_1)m + \frac{a_0}{2} + \frac{a_2}{2}\varepsilon_2$$

Pair correlation

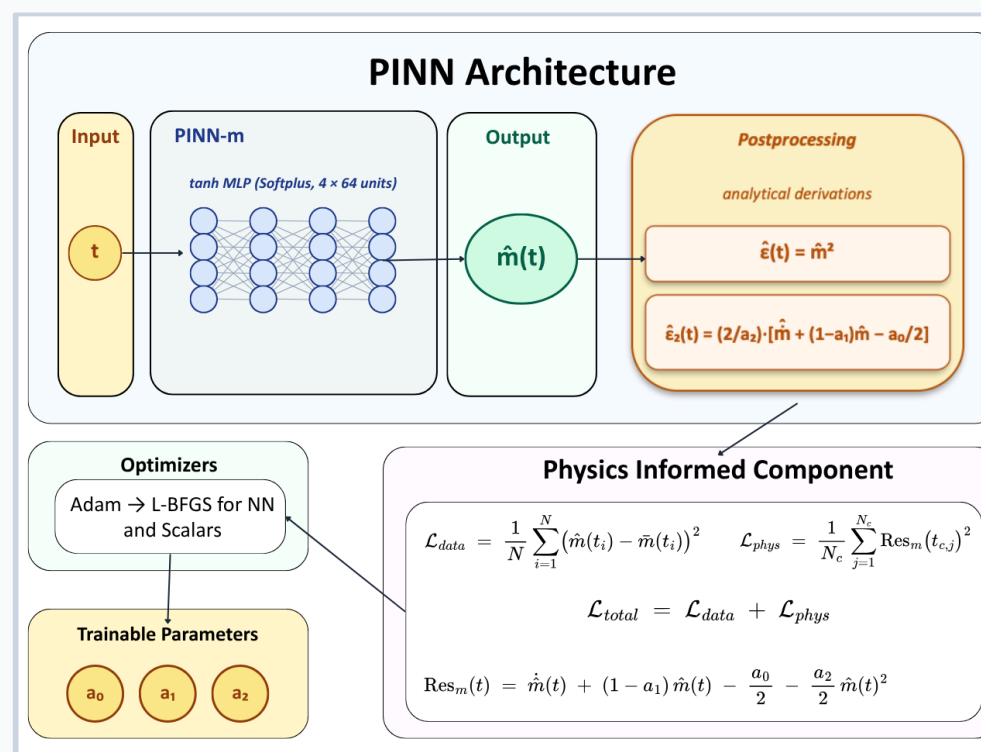
$$\dot{\varepsilon} = -2\varepsilon + a_1 + (a_0 + a_2)m + a_1\varepsilon_2$$

Independent Spins

$$\hat{\varepsilon}_2(t) = \frac{2}{a_2} \left[\dot{\hat{m}}(t) + (1 - a_1)\hat{m}(t) - \frac{a_0}{2} \right]$$

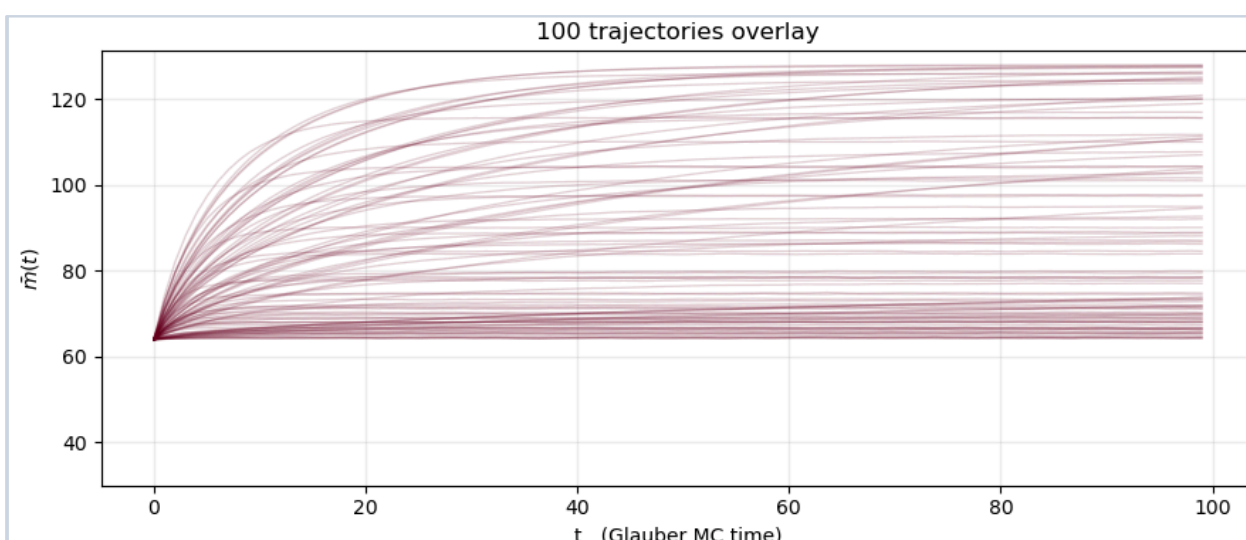
PINN Architecture and Losses

MLP (4 layers, 64 Units) maps time to $m(t)$, with three tanh-bounded scalars (a_0, a_1, a_2) using SoftPlus activation, followed by derived ε . Training combines an MSE data loss with a physics loss on the m-equation residual, optimized by Adam then L-BFGS. The result is a small, fast surrogate for the chain's full dynamics that obeys the physics.



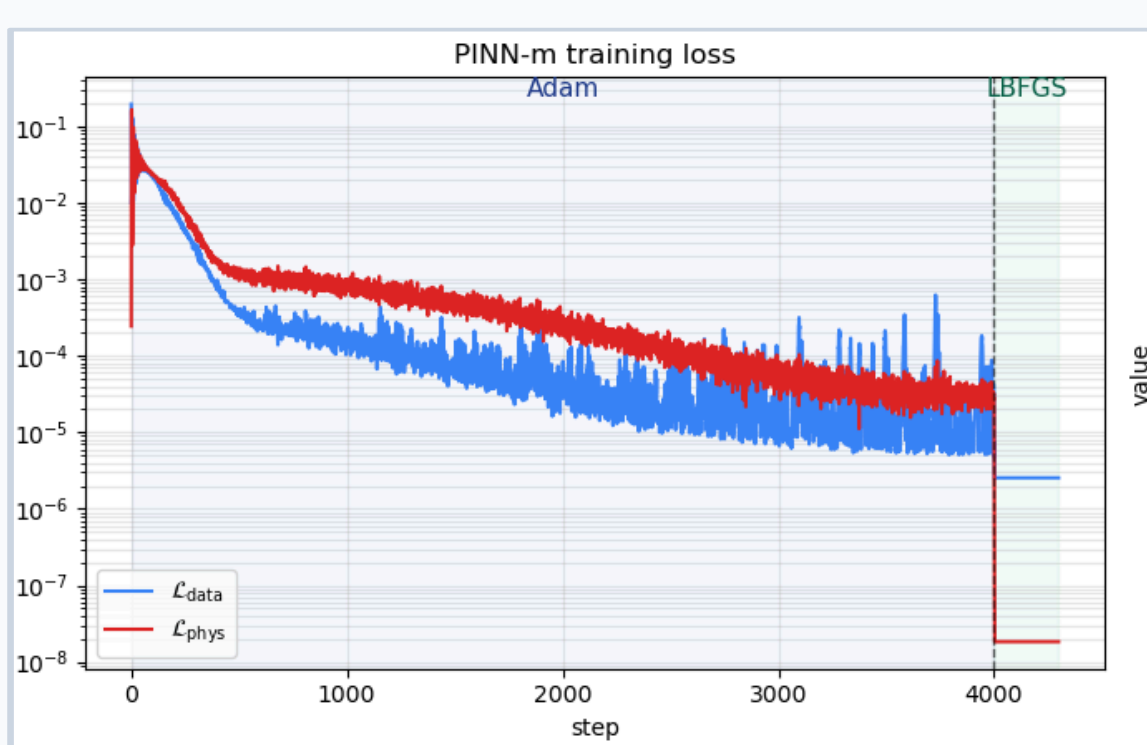
Data Generation

Glauber Monte Carlo simulations of a 1D Ising chain (256 spins, periodic boundaries) provide the data. A 100-condition grid in (β, h, J) is swept, with 8 independent realizations averaged per condition to give smooth $\bar{m}(t)$. The simulator is JIT-compiled with Numba for speed.

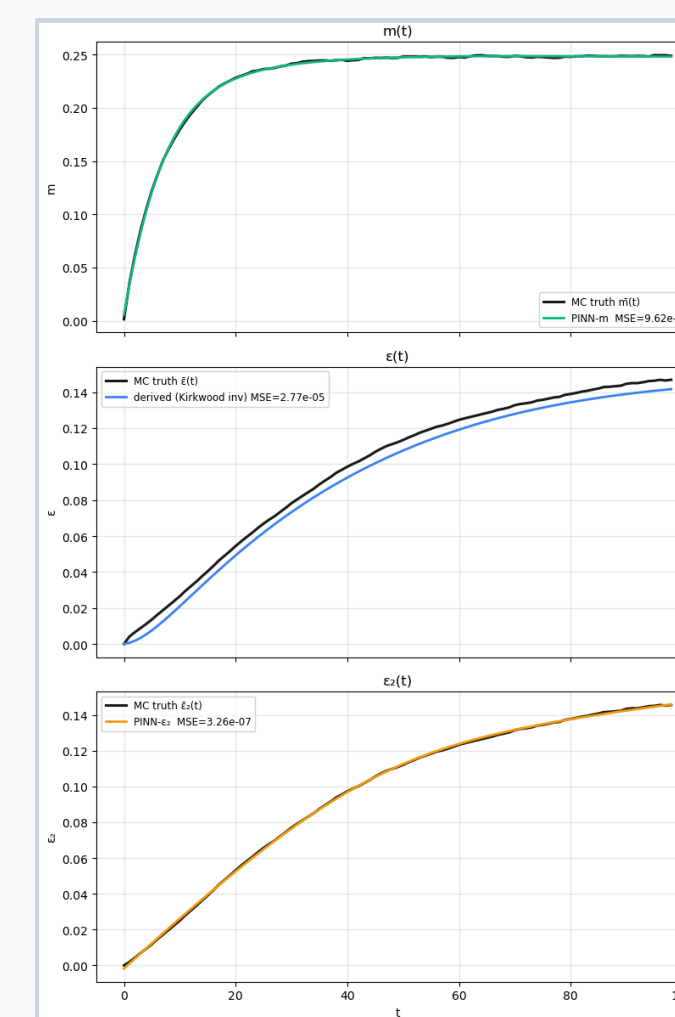


Training Curves

Adam handles the bulk of the descent. L-BFGS then drops the loss by several more orders of magnitude, with data and physics terms converging together rather than trading off.



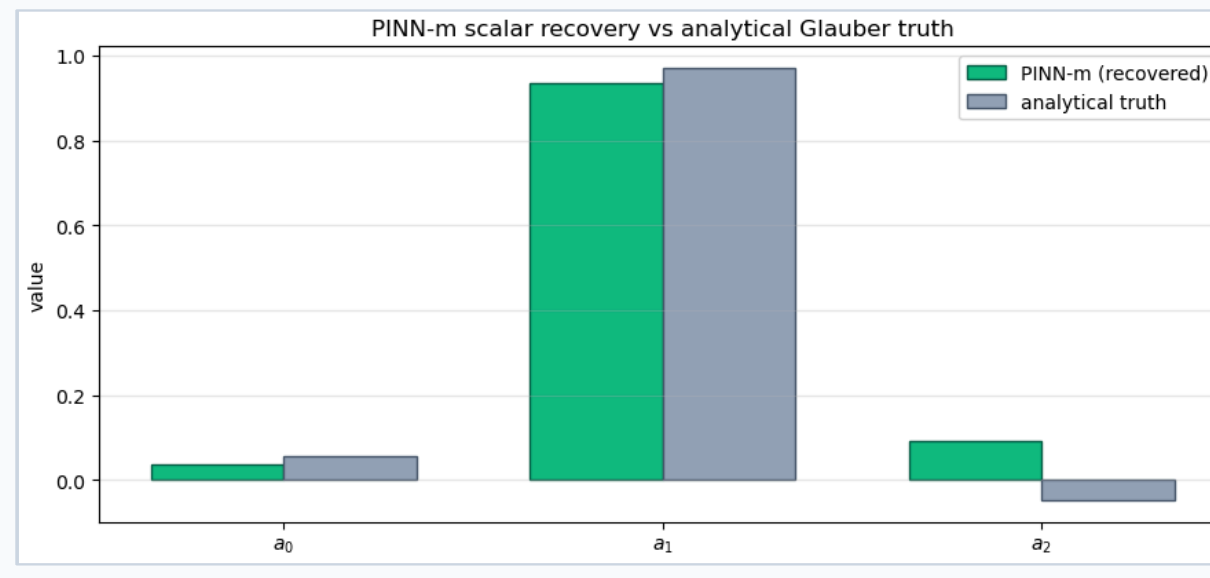
Data Fitting



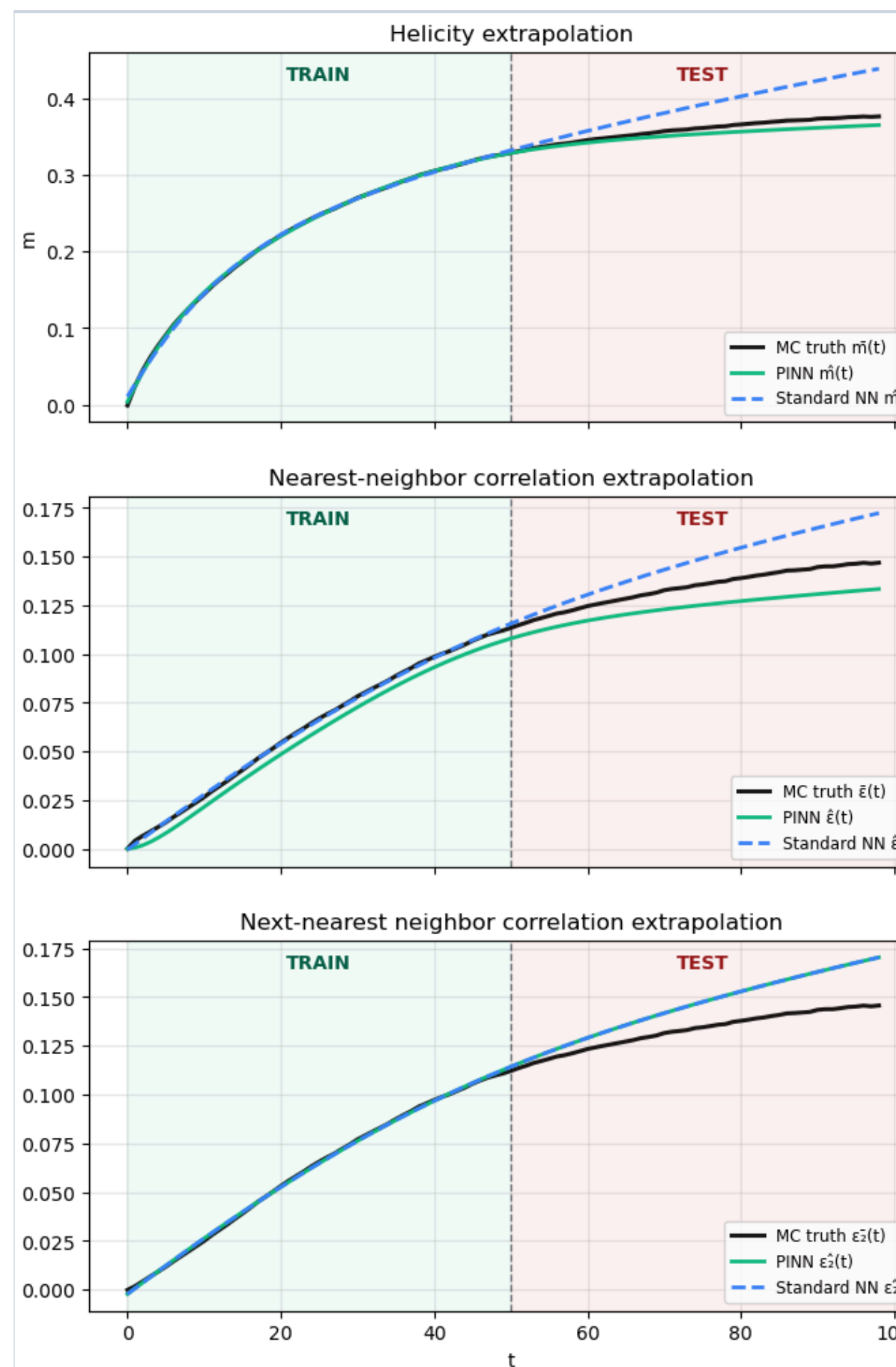
The figure summarises the in-domain fit: $m(t)$ and $\varepsilon_2(t)$ essentially are indistinguishable from the MC truth, and $\varepsilon(t)$ tracks it with a small constant offset. The result is a free check on the physics: the network was trained on m alone, but correlators it predicts still track the simulator.

Scalar Predictions

The PINN recovers all three Glauber coefficients from $m(t)$ alone. The dominant a_0 and a_1 matches the analytical truth almost exactly, while a_2 falls within the expected margin



Comparison with Standard NN



Conclusions & Future Work

- The PINN reproduces all three chain observables from a single helicity trajectory, with MSE of 10^{-5} or better
- The Glauber coefficients (a_0, a_1, a_2) are recovered as a result of training
- Nearest and Next-nearest neighbor correlations are derived analytically from helicity and stay consistent with the MC
- Successful Prediction of 1D Ising System Dynamics
- Future work: apply model to real experimental polymer melting data and consider more complex models

References

Zimm & Bragg, J. Chem. Phys. (1959); Stepanyan et al. J. Stat. Mech. 033202 (2024); Landau & Lifshitz, Stat. Phys. Vol. 5; Raissi et al.